

Pose-Only Edge Action Sensing with Enforced Volatile Memory Confinement

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Abstract—Real-time vision pipelines often process full RGB frames, even though downstream inference uses only geometric information. This design choice results in unnecessary high-resolution biometric texture in intermediate computations. Though the final decision logic utilizes only sparse spatial structure, the processing of RGB frames still has to deal with high-resolution biometric texture. This work introduces a pose-only sensing architecture for inference at the very edge under strictly bounded volatile memory constraints. Instead of storing texture-rich RGB matrices for subsequent perceptual processing, the perception layer (a) extracts sparser skeletal coordinates and (b) reuses the same frame buffer for the next incoming frame within a limited time window (e.g. $\Delta T_{\text{volatile}} \leq 33$ ms). A rank-based projection analysis then shows that the skeleton-reconstruction problem has insufficient degrees of freedom for uniquely reconstructing the original pixel space under certain adversarial settings. The P3P algorithm estimates the camera center via the Perspective-n-Point (PnP) paradigm while coordinate uncertainty is reduced using a confidence-adaptive Kalman filter with $R_k = \sigma^2 / (c_i + \epsilon)$. Across 1,200 labeled kinematic events, the perception layer achieves 97.4% detection accuracy (Precision: 0.98 ± 0.01 , Recall: 0.96 ± 0.02) while operating at under 33 ms end-to-end latency. These results suggest that geometric edge analytics can be made to operate within strictly limited exposure windows by jointly considering dimensional projection and memory constraints as co-design dimensions rather than separate optimization axes.

Keywords—Constrained Edge Perception, Geometric Abstraction, Volatile Memory Confinement, Dimensional Projection, Pose Estimation, Bounded Exposure Sensing.

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INTRODUCTION

The main thesis of this paper is that the problem of persistent exposure of biometrics can be solved by introducing a bounded volatile-memory confinement with sparse geometric projection, so that privacy violations are minimized in the context of visual inferences at the edge. This paper addresses the issue strictly within the confines of the perception systems, focusing on the geometric aspect, not the privacy governance aspect, leaving the latter to the specific applications.

Representational Excess in Conventional Vision Pipelines

Many vision tasks on the edge only use spatial relationships, but traditional vision pipelines decode full RGB frames. Dense RGB tensors carry much more biometric information than needed for downstream tasks, increasing the attack surface of the pipeline unwisely. Geometric abstraction can satisfy kinematic inference requirements without preserving identity-bearing texture. The core issue is representational mismatch, not merely storage duration: capturing detailed texture data far exceeds what kinematic event detection requires [Ref].

Subsystem Scope and Canonical Separation

Within the ScholarMaster architecture, this subsystem performs geometric projection, constrains volatile optical exposure, and emits sparse coordinate abstractions. This subsystem does NOT perform

identity retrieval (Paper 7), orchestrate activation governance (Paper 9), manage cryptographic provenance (Paper 8), derive runtime verification theorems (Paper 18), or define irreversible governance doctrine (Paper 17).

Geometric Abstraction and Volatile Confinement

To address the representational mismatch, the perception layer projects dense optical input into sparse geometric structure and constrains raw-frame lifetime. A pose estimation network operating on edge hardware extracts sparse coordinate representation, after which the raw frame buffer is explicitly overwritten. Memory lifetime itself acts as a security boundary.

This approach is structurally different to post-hoc protection mechanisms, such as blinding, masking, or addition of noise [Ref], which act on retained data. Here, exposure is reduced by limiting lifetime rather than by post-processing obfuscation; zeroization bounds the risk of persistence, and operational constraints are applied to confinement boundaries, which can be measured.

The core technical elements include:

- A formally bounded volatile memory window enforced through OS-level mechanisms.
- A rank-based dimensionality analysis demonstrating structural underdetermination of the pose projection.
- A rule-based geometric event classifier stabilized via confidence-adaptive Kalman filtering and classical PnP orientation estimation.

RELATED WORK

Topological and Geometric Sensing

Existing approaches to exposure-constrained sensing broadly fall into two categories – those that abstract away hardware-specific details, and those that perform post-processing anonymization [Ref]. Depth-domain approaches capture a distance field rather than an RGB texture, which limits the biometric surface resolution available for reconstruction.

Although active depth sensors do have the advantages listed above, they also have several significant limitations. Most notably, these sensors are often expensive, power-hungry, and have a limited useful range, making them impractical in many contexts (such as lecture halls or on trains). The method presented here performs a similar task to topological sensing, but using standard RGB cameras, and aggressively reduces the dimensionality of the problem at the software level [Ref].

Ryoo et al. [Ref] showed results of activity recognition from extremely low resolution imagery (e.g. 16x12). While this involves loss of high-level facial features, the matrix of lower resolution is still a matrix, still open to upsampling by generative algorithms. Recent progress in GANs and diffusion models show that even a degraded pixel-based input could be utilized to synthesize realistic imagery. The low-resolution skeletal vector, if used as input, would bypass the matrix inversion process entirely, replacing it with a constrained coordinate synthesis process.

Lightweight Pose Estimation for Edge Deployment

Pose estimation algorithms have been transitioned from computationally expensive bottom-up approaches such as OpenPose [Ref] to more efficient top-down solutions such as YOLOv8-pose [Ref], BlazePose [Ref] and ViTPose [Ref], which enable real-time performance (>30 FPS) on embedded systems via depthwise separable convolutions and efficient anchor-based design.

Most of the prior work focuses on the improvement of the accuracy of detection (e.g. mAP on COCO [Ref]). In contrast, we shift the attention to system-level containment: the pose network is viewed as a projection operator, whose output is used as geometry information, while its input buffer is deallocated right after inference. Our contribution is primarily related to the memory footprint around the

inference stage, not the accuracy of pose estimation.

Head Pose Estimation Without Texture

The head orientation is commonly estimated using a deep regression model taking cropped face images as an input [Ref]. This approach requires a large amount of textured information that cannot be used within the current framework.

Instead, we formulate the problem in the classical Perspective-n-Point (PnP) [Ref] formulation. In particular, we utilize sparse 2D landmarks projected onto a generic 3D cranial model to reconstruct the 6-DoF orientation, which does not require analysis of facial texture, thus complying with the volatile memory constraints.

SENSING CONSTRAINTS AND THREAT MODELING

Signal Isolation Objective

we first define the signal-separation objective of the sensing pipeline. In conventional pipelines, these signals are inextricably linked: the same RGB tensor carries both geometric structure and identity-bearing detail.

The perception pipeline introduced here enforces a deliberate separation (isolating spatial topology from textural features). The pose estimator acts as a compressive transformation, allowing only low-frequency spatial topology (joint coordinates) to pass downstream. High-frequency components—skin texture, pigmentation gradients, microstructure—are discarded prior to persistence. In signal-processing terms, the pipeline behaves like a spatial abstraction filter. It transmits coordinate relationships while eliminating pixel-level content.

Adversary Model and Operating Assumptions

We adopt a constrained but realistic adversarial model. The adversary may obtain access to persistent logs or stored coordinate events and may perform retrospective forensic analysis. However, the attacker does not possess real-time hardware extraction capability during the brief frame-processing interval.

The following assumptions define the Trusted Computing Base (TCB):

- No active Direct Memory Access (DMA) interception during live processing [Ref].
- No physical cold-boot attack against system RAM [Ref].
- No kernel-level compromise capable of intercepting memory allocations at the microsecond scale during frame processing.

Under these assumptions, the dominant risk shifts from live interception to retrospective harvesting. The mitigation strategy therefore focuses on ensuring that raw optical data never persist beyond a tightly bounded processing window. Dimensionality reduction also helps constrain inversion risk by structurally collapsing the signal space.

Formal Volatile Confinement Window

The confinement model treats raw optical frames as transient computational artifacts rather than persistent data objects. We define the relevant timestamps:

- T_{capture} : frame arrival into kernel space
- $T_{\text{vectorize}}$: completion of pose inference
- T_{wipe} : completion of explicit zeroization

The existence interval of the raw frame is therefore:

$$\text{Equation: } T_{\text{capture}} < t < T_{\text{wipe}}$$

The volatile exposure duration is defined as:

$$\text{Equation: } \Delta T_{\text{volatile}} = T_{\text{wipe}} - T_{\text{capture}}$$

To prevent queue accumulation or swap-induced persistence, the pipeline constrains:

$$\text{Equation: } \Delta T_{\text{volatile}} \leq 33\text{ms}$$

This bound is compatible with a 30 FPS real-time processing constraint. This is achieved by overwriting raw frames before later captures take place, thus making the exposure window both measurable and repeatable. With the given adversary model, such a procedure renders reconstruction from the storage device impossible since no raw frame remains after this exposure window has passed.

SIGNAL PROCESSING METHODOLOGY

One-Way Pipeline Structure

From an architectural perspective, the sensing pipeline enforces a strict one-way projection boundary, elevating vectorization as an architectural irreversibility boundary rather than merely a preprocessing stage. Raw optical data flows through a volatile region, undergoes vectorization, and is explicitly destroyed. Downstream systems never regain access to dense optical tensors; coordinate emission is intentionally lossy, and persistence occurs only after projection and destruction. Only the resulting coordinate vectors are permitted to enter the persistent evaluation stage—typically 34 scalar values under the COCO convention.

There are no feedback paths or secondary buffers capable of reintroducing raw pixels into downstream components. This structural isolation prevents the reconstitution of texture-space information.

[FIGURE: Architectural Pipeline / Validation Curves]

\caption{The Sensing Pipeline. Data flows horizontally through the volatile layer where it is vectorized and explicitly wiped. Only abstract coordinate data descends to the persistent evaluation layer.}

\label{fig:arch}

\end{figure}

Topological Event Formalization

The pose network outputs joint coordinates (x_i, y_i) and confidence scores c_i for $K=17$ keypoints under the COCO convention [Ref]. For example:

$$\text{Equation: } W_R = [x_{\{16\}}, y_{\{16\}}], \quad E_R = [x_{\{8\}}, y_{\{8\}}]$$

To reduce noise propagation, coordinates are accepted only if:

$$\text{Equation: } \text{Valid}(W_R) = \mathbb{I}\{c_{\{16\}} > \tau_c\}$$

where $\tau_c = 0.65$.

Because subjects may stand at varying depths relative to the camera, thresholds therefore scale with apparent body size. We define:

$$\text{Equation: } D_{\text{shoulder}} = \sqrt{(x_6 - x_5)^2 + (y_6 - y_5)^2}$$

An elevation event is triggered when:

$$\text{Equation: } H_{\text{event}}(t) = \mathbb{I}\{y_{\{W_R\}} < y_{\{E_R\}} - \alpha_{D_{\text{shoulder}}}\}$$

This relative formulation removes dependence on absolute pixel location and improves robustness across camera distances.

Rank-Based Projection and Structural Underdetermination

To evaluate reconstruction feasibility, we treat the pose extractor as a non-linear projection:

$$\text{Equation: } f: \mathbb{R}^{37632} \rightarrow \mathbb{R}^{34}$$

A 112x112 RGB crop resides in $\mathbb{R}^{37632}$. The skeletal output contains only 34 scalars.

The Jacobian rank satisfies:

$$\text{Equation: } \text{Rank}(J_f) \leq 34$$

By the Rank-Nullity Theorem:

$$\text{Equation: } \dim(\ker(f)) \geq 37,598$$

Underdetermination arises from this basic dimensional collapse, but not from adversarial inability. The geometric structure is retained while texture-space information is discarded. Even given knowledge of the projection mapping, the null-space ambiguity renders an astronomically large set of input vectors compatible with any given output.

While generative models could synthesize a likely human pose given the skeleton, the deterministic reconstruction of the original high frequency biometric texture is not uniquely solved without additional constraints [Ref]. The transformation does not hide identity carrying details but rather discards them, structurally.

Perspective-n-Point (PnP) Orientation Estimation

The authors calculate the head orientation using the classical perspective-n-point (PnP) geometry. The paper notes that the geometric approach stabilizes perception, if the certainty of individual measurements is bounded, rather than making semantic assumptions. Finding the orientation becomes possible without processing dense

pixel patterns, by relying on sparse 2D landmarks projected onto a 3D cranial model:

$$\text{Equation: } \begin{bmatrix} u \\ v \end{bmatrix} = K \begin{bmatrix} R & t \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

The rotation matrix R and translation vector t are obtained by minimizing reprojection error:

$$\text{Equation: } \min_{\{R, t\}} \sum_{i=1}^n \|u_i - \hat{u}_i\|^2$$

where $\hat{u}_i = \pi(K, R, t, X_i)$.

In evaluation, mean reprojection error remained below 2.5 pixels. Euler angles are extracted:

$$\begin{aligned} \psi &= \arctan2(R_{21}, R_{11}) \\ \theta &= \arctan2(-R_{31}, \sqrt{R_{32}^2 + R_{33}^2}) \\ \phi &= \arctan2(R_{32}, R_{33}) \end{aligned}$$

Orientation is therefore inferred entirely from geometry, without referencing facial texture, stabilizing inference.

Confidence-Adaptive Kalman Filtering

Edge-based pose estimators can exhibit high-frequency jitter, especially under occlusion. We model joint dynamics as:

$$\text{Equation: } x_k = [x, y, v_x, v_y]^T$$

Prediction:

$$\text{Equation: } x_{k|k-1} = F_k x_{k-1|k-1}$$

Covariance propagation:

$$\text{Equation: } P_{k|k-1} = F_k P_{k-1|k-1} F_k^T + Q_k$$

Kalman gain:

$$\text{Equation: } K_k = P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1}$$

Instead of a fixed measurement covariance, we define:

$$\text{Equation: } R_k = \frac{\sigma^2}{c_i + \epsilon} \cdot I$$

Lower confidence values increase R_k , causing the filter to discount unreliable observations. High confidence reduces R_k , allowing rapid correction. The update step remains:

$$\text{Equation: } x_{k|k} = x_{k|k-1} + K_k y_k$$

A finite-state persistence constraint further filters transient motion:

$$\text{Equation: } \text{Valid}(t) = \sum_{i=t}^{t+\Delta} H_{\text{event}}(i) \cdot \Delta > 1.0$$

This enforces temporal stability before emitting an event.

MEMORY-AWARE SENSING CONSTRAINTS

Memory lifetime itself acts as a security boundary. Maintaining a bounded exposure window requires coordination between application logic and the underlying operating system memory manager. Without explicit safeguards, operating systems may retain sensitive frame data within swap space, cache hierarchies, or freed memory regions [Ref]. The confinement boundaries are operationally measurable and depend heavily on disciplined memory management rather than application-level logic alone.

OS-Level Page Locking

Modern operating systems rely on virtual memory paging. Under memory pressure, embedded-memory pressure conditions, inactive pages may be written to persistent storage (e.g., Linux swap

partitions). If raw RGB buffers were paged out before zeroization, the confinement boundary would be breached.

To prevent this, allocated frame buffers are pinned in physical memory using POSIX `mlock()`. This prohibits the kernel's memory manager from swapping the pages to disk. The lock is established during initialization and released only after secure zeroization. By ensuring that frame buffers remain resident in RAM throughout their lifetime, the pipeline reduces disk-level persistence risk under assumptions.

Secure Zeroization and Compiler Behavior

Deallocation via `free()` or `delete` does not overwrite memory. The underlying physical pages may retain intact pixel data until reused by another process [Ref]. Consequently, explicit zeroization is mandatory.

nevertheless, using `memset()` naively is not sufficient. Modern compilers can optimize away redundant memory writes as a part of their Dead Store Elimination (DSE) optimization passes. In particular, during the initial testing of the idea, builds done with clang -O2 optimized out the `memset()` call when the buffer was freed immediately after, making the memory content still available for inspection.

To guarantee the execution of the overwrite, we use `explicit_bzero()` or `memset_s()` secure functions which are resistant to DSE, and cast pointers to volatile to prevent the compiler optimizations. In addition, if available on the platform, we also use instructions to flush the data from the CPU caches (`_mm_clflush`). By doing so, we cannot guarantee that there will be no remnants of the data anywhere in the microarchitecture, but the probability of that happening becomes extremely low.

The combination of page locking, explicit zeroization, and optional cache flush primitives renders the volatile boundary operationally measurable rather than theoretical.

ALGORITHM DESIGN

The pipeline is divided into two clearly separated phases: a volatile extraction stage and a persistent logical stage. The separation is structural rather than conceptual; raw pixel data is destroyed before any event emission occurs.

Algorithm 1 reflects this ordering:

- Pose inference is performed directly on the volatile frame buffer.
- The buffer is securely overwritten.

- Filtered coordinate vectors are processed by geometric logic.
- Only abstract event records are appended to persistent storage.

[FIGURE: Architectural Pipeline / Validation Curves]

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The skeletal overlay shown in Figure \ref{fig:skeleton} illustrates the abstraction level retained by the perception logic.

Notably, the secure deletion routine executes immediately after vectorization. This sequencing ensures that even if downstream logic stalls, raw frames are not retained in memory.

EXPERIMENTAL EVALUATION

Dataset and Evaluation Conditions

Experiments were performed on a data set of 1,200 manually labeled kinematic events collected across a range of illumination and occlusion conditions. This is a standard procedure for evaluating a method’s ability to detect geometry-based events (independent of identity or texture-based analysis) against the background of potential threats (illumination/occlusion). The five-fold cross-validation procedure was used to decrease variance.

Geometric Event Detection Performance

Across all scenarios, precision remains high. Under ideal lighting, performance approaches near-saturation levels ($F1 = 0.97 \pm 0.01$). Performance degradation under low light and partial occlusion primarily affects recall rather than precision.

TABLE: Geometric Detection Performance Metrics (N=1,200)

This behavior is explained by the conservative nature of the geometric logic: when confidence in a prediction decreases, the adaptive Kalman filter reduces exposure to false positives. The resulting precision and recall figures (Precision: 0.96 ± 0.01 ; Recall: 0.90 ± 0.02) demonstrate that, on average, the system performed as intended during tests with limited exposure.

Latency Decomposition and Confinement Verification

An important metric is total end-to-end latency relative to the confinement bound.

TABLE: Latency Decomposition (Edge SoC)

The zeroization overhead (~ 1.2 ms) is a deliberately accepted compromise between performance and memory hygiene; attempts to offload the wipe to a parallel thread yielded only minor improvements, indicating that secure erasure is not a bottleneck under the given configuration.

Crucially, experimental evaluation indicates ~ 30 FPS throughout sustained testing, confirming that the confinement constraint is operationally viable.

[FIGURE: Architectural Pipeline / Validation Curves]

Signal Stabilization Ablation Study

To quantify the contribution of the confidence-adaptive Kalman filter, we evaluated 500 challenging sequences exhibiting moderate occlusion.

- False Positive (FP) rates:
- No filter: 14.5%

- Standard Kalman filter: 8.2%
- Confidence-adaptive Kalman + FSM: 1.5%

The adaption of R_k according to network confidence was critical to avoid false detections due to jitter. Otherwise, the filter cannot differentiate between true motion and bounding-box instability.

DISCUSSION

Residual Behavioral Signatures

Although the pipeline does not encode pixel persistence at all, geometric signals can potentially carry behavioral persistence over sufficiently long time horizons. Gait, body posture, and characteristic movements can be used as soft biomarkers of identity if an adversary has learned behavioral patterns of a particular individual in advance.

The perception layer therefore reduces the biometric retention surface, but by no means eliminates the risk of identity compromise. Practical measures include periodic refreshing of internally generated tracking tokens, isolation of sessions to prevent trajectory reconstruction across sessions, and a certain degree of spatial aggregation to limit the temporal resolution of behavioral fingerprints. This should be considered risk reduction within a concrete threat model, not an absolute anonymity guarantee.

Geometric Sufficiency for Event Inference

A frequently raised concern in the context of reduced sensing modalities is the potential loss of dimensionality that may compromise operationally relevant information. The results presented here demonstrate that, for the purposes of kinematic event detection (spatial gestures, transitions in pose/orientation), the information loss in the projection from a fully descriptive 3D mesh representation is minimal. Information contained in the skin geometry and surface texture is largely irrelevant to the event recognition task.

This observation has implications for the design of perceptual systems beyond the scope of the current work. Specifically, this result demonstrates that many common classes of institutional sensing tasks may be performed using only geometrically encoded information while avoiding the memory and computational burdens of full-texture retargeting. The obvious caveat is that fine-grained facial affect analysis, lip-reading, or expression-classification tasks would benefit substantially from the retained high-frequency texture information. This architecture is explicitly designed for geometric event detection and not general-purpose visual analytics.

Limitations

All of the geometry in the current implementation is calculated from monocular 2D pose vectors, which is why occlusion instability and scheduler latency skewer both the accuracy of orientation and the reliability of event detection. A 2D projected skeleton suffers from an inherent inability to distinguish between raised and outstretched limbs at an angle, which is a problem for depth-based estimation of 3D joints from camera space.

The memory confinement model takes as its Trusted Computing Base (TCB) an environment that does not allow real-time DMA interception, cold-boot attacks, or kernel-level memory snooping. The assumptions are not necessarily safe as the system cannot withstand privileged kernel compromises or more general hardware-level attackers. While cache-line residue can be eliminated by using specific CPU primitives, such protections are not available for all CPUs and cannot be used in all environments. Moreover, there are no guarantees of hardware-rooted attestation at lower levels.

Residual gait-signature leakage and soft biometric leakage via motion patterns are problems that cannot be solved by the geometric projection. The confinement strategy helps reduce texture exposure, but it does not eliminate behavioral correlation given enough observational time by a determined attacker. The future research should focus on finding a lightweight 3D estimation model that could mitigate depth ambiguity while still operating within the 33 ms confinement budget.

CONCLUSION

This constrained geometry perception study evaluated the viability of achieving real-time geometric event sensing under limited temporal exposure of raw optical information. The system time constraints were enforced by operating system (OS)-level page locking and secure zeroization procedures, thus, restricting volatile memory exposure within a 33 ms time window (Δ_{volatile}).

The rank-based projection analysis reveals that a sparse geometric projection produces an extremely large collapse in dimensionality ($\mathbb{R}^{37632} \rightarrow \mathbb{R}^{34}$), thereby rendering the reconstruction of the projected high-dimensional pixel space under the given threat model not structurally determined. A structurally undetermined reconstruction possibility is a geometric characteristic of the projection, rather than a claim about impossibility against an adversary.

Experimental results show that the combination of pose-based dimensional reduction, confidence-adaptive Kalman filtering, and geometric PnP orientation yields operationally stable event detection ($F1: 0.93 \pm 0.01$) within the confinement bound. The possibility of residual gait-signature leakage, monocular depth ambiguity, and TCB assumptions are all disclaimed. With respect to the ScholarMaster architecture at large, this capability is strictly limited to geometric projection and the emission of sparse coordinate abstractions, devoid of identity retrieval functions (Paper 7) and orchestration governance (Paper 9).

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